

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Design and Analysis of Algorithms | Course Code: 21CS42 | Credits: 4

Exam Session: End Semester Exam 2023

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Solve the 0/1 Knapsack Problem using Dynamic Programming for capacity $W = 5$ and items: Item 1 (weight 2, value 12), Item 2 (weight 1, value 10), Item 3 (weight 3, value 20), Item 4 (weight 2, value 15). Build the complete DP table and find selected items. [8 Marks]

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Q2. State Master Theorem for divide-and-conquer recurrences $T(n) = a T(n/b) + f(n)$. Solve: (i) $T(n) = 4T(n/2) + n$, (ii) $T(n) = 2T(n/2) + n \log n$, (iii) $T(n) = 3T(n/4) + n^2$. [8 Marks]